DINOJAN VIVEKANANTHARASA

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PROFESSIONAL SUMMARY

A Computer Engineering graduate focused on **applied AI systems and machine learning**, with one year of professional full-stack experience shipping production platforms. Experienced in building LLM agents and RAG pipelines, fine-tuning open models with QLoRA, evaluating them against baselines, and taking models to production with MLflow, Docker, and Kubernetes.

EDUCATION

BSc.Eng (Honours) in Computer Engineering **Feb 2022 – Sep 2026**
Faculty of Engineering, University of Ruhuna, Galle, Sri Lanka

- **Certificate Course in Soft Skills Development** (2022–2024): communication, leadership, problem solving

GCE Advanced Level **Oct 2020**
V/Vavuniya Tamil Madhya Maha Vidyalayam *Physical Science: A, C, S*

PROFESSIONAL EXPERIENCE

Yarl Ventures (PVT) Ltd, Jaffna **May 2025 – May 2026**
Full Stack Developer *Dec 2025 – May 2026*

- Built a read-only viewer-admin role in MediMan (API permissions, route protection, role-based dashboard), plus clinic-management and appointment-rescheduling APIs and UI.
- Contributed to EDUS LMS, a subscription platform with monthly billing: payments, wallets, chat, role-permission testing, and API optimisation.

Full Stack Developer Intern *May 2025 – Nov 2025*

- Built the MediMan admin panel, adding server-side pagination and field projection to cut API payload sizes.
- Integrated Nodemailer, Twilio SMS/OTP, and Firebase push; debugged production via AWS CloudWatch logs.

TECHNICAL SKILLS

Programming: Python, SQL, TypeScript, Java

AI/ML: LLM Agents, RAG, QLoRA Fine-tuning, PyTorch, TensorFlow, Scikit-learn

MLOps & Tools: LangChain, Hugging Face, FastAPI, MLflow, Docker, Kubernetes (kind)

Soft Skills: Teamwork, Problem Solving, Time Management

PROJECTS

LLM Research Agents **Sep 2026 – Oct 2026**
Python, LangChain, Groq, Pydantic, Phi-3-mini (QLoRA), Hugging Face PEFT/TRL

- Financial research system: tool-calling agents plus a QLoRA fine-tuned Phi-3-mini risk extractor.
- Fine-tune raised schema-valid JSON from 1/12 to 12/12 and ROUGE-L by 31% over the base model.

Credit Default MLOps Pipeline **Oct 2026**
Python, LightGBM, Optuna, MLflow, SHAP, FastAPI, Docker, Kubernetes (kind), GitHub Actions, Evidently

- End-to-end MLOps pipeline: LightGBM tracked in MLflow, served by FastAPI on Kubernetes (kind).
- Test ROC-AUC 0.79; retraining promotes a model only if it beats the champion, with drift alerts.

Larkway – Policy-Grounded AI Support Agent **Aug 2026 – Oct 2026**
Python, FastAPI, Claude / Groq / Gemini APIs, LLM tool calling, Pydantic, Vercel

- Live AI support agent with 9 tools over policies, contracts, and order data, also served over MCP.
- A deterministic policy engine, not the LLM, computes fees and SLAs; 156 tests and 16 evals pass.

TRACES – Post-Quantum Secure Vehicular Network Defense **Dec 2025 – Sep 2026**
Python, NS-3, Qwen2.5-7B (QLoRA), Hyperledger Fabric
- Final-year research: built the LLM cascade, NS-3 simulation, and post-quantum and Fabric layers.
- Detects 7 attack classes at 0.867 macro-F1 on held-out data, with a detector running in about 82 μs a window.

AI-Powered Financial Document Q&A System (RAG) **Jan 2026 – May 2026**
Python, FastAPI, Streamlit, ChromaDB, Groq Llama 3.3 70B
- RAG app grounding Llama 3.3 70B in 10-K and SEC filings, citing the source page of every answer.
- Hit@1 retrieval of 1.000 and 84% keyword recovery, against under 1% for an ungrounded LLM; live.

AutoGradePro – LLM-Assisted Assessment Marking **Jun 2024 – May 2025**
Next.js, TypeScript, Django, PostgreSQL, Ollama (Qwen2.5), rule-based checks
- Grades short answers on meaning with a local LLM; contributed to its frontend, backend, and accuracy tests.
- Published accuracy: 100% single-word, 92.9% list, 84.6% numerical; its client tested it on 40+ scripts.

ECG Image Classification **Dec 2024 – Apr 2025**
Python, TensorFlow/Keras, CNN, FastAPI
- Built a transfer-learning CNN pipeline for ECG images and its FastAPI inference service.
- 95% test accuracy on 24,799 test images (MIT-BIH, PTB), served via single and batch prediction APIs.

PUBLICATIONS

G. S. J. Venujan, M. F. A. Munsif, **V. Dinojan**, T. Jegan, and G. G. N. Sandamali, "AutoGradePro: An Automated Grading System for Short-Answer Assessments Using Rule-Based Methods and Large Language Models," in *Proceedings of the Vice Chancellor's Awards and Academic Sessions 2026*, University of Ruhuna, Sri Lanka, Mar. 2026, Abstract ABS-064, p. 35. ISSN 2362-0412.

VOLUNTEER EXPERIENCE

Member **Apr 2024 – Present**
IESL, Galle

Member **Sep 2023 – Dec 2024**
IEEE Student Branch, Ruhuna, Galle

Treasurer **Apr 2023 – Apr 2024**
All University Students' Development Association, Vavuniya

Senior Prefect **Jul 2017 – Jul 2018**
V/V.T.M.M.V, Vavuniya

ACHIEVEMENTS

- Participated in **IEEE Xtreme 18.0 (2024)**, a global 24-hour algorithmic coding contest (3-member team)
- Competed in **IEEE INSL Southern Provincial Competition**, 30th October 2024
- **3rd Runner-Up**, Provincial Level Under-19 Boys Chess, 2015 and 2016
- Participated in the **Galle Division FIDE 100 Commemorative Chess Event** (Open Category), 2024

REFEREES

- **Dr. Kushan Sudheera**
  Senior Lecturer
  Department of Electrical and Information Engineering, Faculty of Engineering, University of Ruhuna
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- **Dr. Nadeesha Sandamali**
  Senior Lecturer
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